



Research article

Evaluating the cumulative and time-lag effects of drought on grassland vegetation: A case study in the Chinese Loess Plateau

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ABSTRACT

The increased frequency of drought events in recent years is known to be responsible for significantly altering plant biodiversity in many of Earth's ecosystems, though the specifics of vegetation-drought interactions, especially the cumulative and time-lag responses, remains unclear. This study aimed to quantitatively investigate how grassland vegetation over the Chinese Loess Plateau (CLP) reacts to drought, specifically the observed cumulative and time-lag effects which are caused, using a combination of the Normalized Difference Vegetation Index (NDVI) and a multiple time-scale drought index (Standardized Precipitation and Evapotranspiration Index, SPEI). Our results revealed that while drought conditions have widespread cumulative impacts on grass growth in the CLP, the time lag effect of drought covered about half of the total area of the CLP. The cumulative effect of drought on grass was found to take place over various time scales, ranging from 5 to 10 months, while the time lag effect occurred within 2–3 months. The different response time of vegetation growth to the cumulative effect of drought in the CLP was found to be highly related to different water conditions. The accumulated months and mean $r_{max-cum}$ both had a significant negative correlation with the mean annual SPEI ($R^2 = 0.90$, $P < 0.001$; $R^2 = 0.70$, $P < 0.001$, respectively). The lagged months and mean $r_{max-lag}$ were also found to be negatively correlated with the mean annual SPEI ($R^2 = 0.547$, $P < 0.05$; $R^2 = 0.785$, $P < 0.01$, respectively).

1. Introduction

Global climate change is projected to increase the frequency and intensity of drought which occur under warmer conditions (Li et al., 2019a; Wang et al., 2018; Zhang et al., 2019). Drought conditions are capable of influencing the normal growth of vegetation and continued drought can even trigger large-scale landscape change through causing vegetation mortality from water stress (Chu et al., 2019; Haslem et al., 2015; Liu et al., 2019; Zhao et al., 2018a). Drought conditions are also one of the main drivers of reductions in net primary production (Zhao and Running, 2010; Zhao and Running, 2011). Numerous studies have investigated the relationship between drought and vegetation growth (Gouveia et al., 2017; Hua et al., 2017; Xu et al., 2018; Zhao et al., 2018b). For example, Vicente-Serrano et al. (2013) evaluated the correlation between Standardized Precipitation Evapotranspiration Index (SPEI) and global land biome by seasonal maximum Pearson correlation,

and showed that more than 70% of global land vegetation has been affected by drought events. Zhang et al. (2017a) evaluated the correlation coefficient between Normalized Difference Vegetation Index (NDVI) and SPEI by annual maximum Pearson correlation, and found that there was a significantly positive correlation with drought effects on vegetation in most regions of China. The effect of drought accompanied by warmer temperature might be expected to produce even greater impacts on vegetation in the future (Cooley et al., 2019).

Understanding how vegetation responds to drought is challenging, as the visible symptoms of drought-related effects on vegetation growth are not immediately evident. Vegetation growth may not primarily be driven by the present water condition, but by early drought events (D'Orangeville et al., 2018; Peng et al., 2019). After a severe drought, it often takes time for plant to repair the damaged root system and regain the growth capacity it had prior to the drought (Anderegg et al., 2012). The time required for the drought injury to occur mainly depends on

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both the environmental conditions at the time and the species of vegetation (Peng et al., 2019; Wen et al., 2019; Wu et al., 2015). Knowing the variability of the plant's response time to drought is critical for efficient management and conservation of vegetation as well as understanding the mechanism of climate-vegetation interactions (Wen et al., 2019; Wu et al., 2015; Zhao et al., 2018a).

An increasing number of studies have examined the time-lag effect of drought events on vegetation growth (Huang et al., 2018; Wu et al., 2015; Wu et al., 2018b). They have shown that drought has a significant time-lag effect on vegetation growth, which can vary among different species and across different ranges (Braswell, 1997; Peng et al., 2019; Vicente-Serrano et al., 2013). Despite the contributions these studies have made in explaining the interactions between drought and vegetation, they have not yet been able to integrate the magnitude and duration of drought (Hoover and Rogers, 2016). Studies on the time-lag effects of drought often focus on only a single point of time, while ignore the water conditions from that point to time of the observation, increasing large uncertainties in examining the variability of the response time drought. Incremental water deficit has been reported to be related to vegetation greenness (Berry and Mackey, 2018; Méndez-Barroso et al., 2009; Xu et al., 2011). Events such as cumulative water deficits can enhance the effects of drought on ecosystems (Ivits et al., 2016; Peng et al., 2019). Time-lag effects indicate the impacts of former drought events happened at a specific point of time on the current vegetation growth, while cumulative effects are based on the time-lag effects and include the dynamics of climatic conditions (e.g., precipitation and evapotranspiration) for a given period of time. Knowing both the time-lag effects and cumulative effects is important for understanding the interactions between drought events and vegetation growth, and also to exert more efficient vegetation management strategies.

The Chinese Loess Plateau (CLP) is located in arid and semiarid regions, and is the one of the most severe areas of soil and water loss in the world (Shi and Shao, 2000; Zhao et al., 2013). Unfortunately, this region is experiencing an increasing tendency towards droughts both duration and severity (Wu et al., 2018a). Grassland is one of the most important ecosystems, as well as plays the key role in the soil and water conservation (Zhou et al., 2010). It is very suitable for grassland growth

in terms of climate and soil conditions in the CLP and more than 40% of the area is covered by grassland (Zhang et al., 2017b; Zheng et al., 2019). However, few studies have examined the impact of cumulative and time-lag effects of drought on grassland vegetation in the CLP.

In this study, we aimed to evaluate the cumulative and time-lag effects of drought on grass growth in the CLP using time-series NDVI data and the SPEI data. Specifically, we aimed to assess the cumulative and time-lag effects of drought on grassland NDVI in the CLP, and to understand how these effects change with the water balance gradients.

2. Materials and methods

2.1. Study area

The CLP (102°54'–114°33'E, 33°43'–41°16'N) is located in north central China. It is bordered by the Taihang Mountains to the east, the Riyue Mountains to the west, the Yinshan Mountains to the north, and the Qinling Mountains to the south and covers a total area of 6.49×10^5 km² (Wang et al., 2012). The climate is typical of the continental monsoon climate zone, and the zonal distribution of climate conditions from the northwest to the southeast is from arid, to semiarid, to semi-humid (Fig. 1). The mean annual precipitation ranges from 100 mm in the northwest to 750 mm in the southeast, and the annual air temperature ranges from 4.3 °C in the northwest and gradually increases southward to 14.3 °C (Zhao et al., 2019). As one of the earliest human settlements, this region is also of great importance for ecological security and natural resources provisions (Zhao et al., 2013). Severe soil and water loss have led to widespread land degradation in this region; during the past decades, great efforts have been made in vegetation restoration to reduce soil erosion and water loss. About 3766.8 km² croplands were replaced by grasslands during 2000–2010 (Zhang et al., 2016b). With global warming and increased human activity, the frequency and severity of droughts has gradually increased, which has important influence on vegetation in the area (Zhang et al., 2016a). Examining how the newly established vegetation responds to drought in the CLP is urgent for long-term water and soil conservation, as well as to support the development of agriculture.

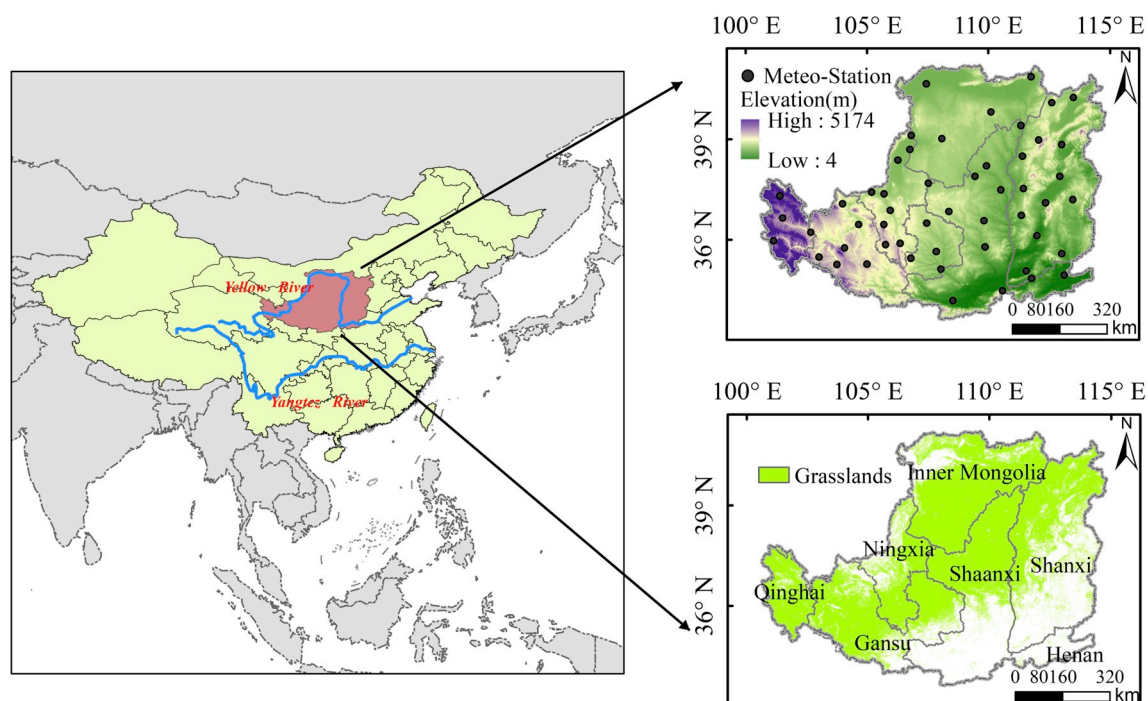


Fig. 1. (a) Geographic location of China Loess Plateau (CLP); (b) Topographic map of the CLP; and (c) Distribution of the grasslands in the CLP.

2.2. Data

Land cover and long-term NDVI data was used to extract the time-series of grassland and SPEI data to assess drought severity. All the spatial datasets used in the study were resampled using a bilinear interpolation algorithm to achieve a final spatial resolution of 8 km.

2.2.1. NDVI time series

The Global Inventory Monitoring and Modelling Systems (GIMMS3g) NDVI dataset was obtained from the National Oceanographic and Atmospheric Administration (NOAA) with an Advanced Very High Resolution Radiometer (AVHRR) to evaluate the variations of grassland vegetation in the CLP (<https://ecocast.arc.nasa.gov/data/pub/gimms/3g.v1/>). This dataset is the longest record of vegetation information, spanning from 1982 to 2015 at a spatial resolution of approximately 8 km × 8 km within a 15-day interval, and has been extensively utilized to monitor vegetation dynamics (Chu et al., 2019; Kong et al., 2018; Li et al., 2019b; Na et al., 2018; Peng et al., 2019; Qu et al., 2018). The maximum value composite (MVC) technique was deployed to generate monthly NDVI.

2.2.2. Land cover data

Grassland distribution in the CLP (Fig. 1) was derived from the 5.1 MODIS land cover product (MCD12Q1) collection downloaded from the USGS (United States Geological Survey) Land Cover Institute (https://landcover.usgs.gov/global_climatology.php) with a spatial resolution of 500 m (Broxton and Zeng, 2014). This dataset is based on the 10-year (2001–2010) land cover type data from Collection 5.1 MCD12Q1 and has been classified into 17 major land cover types (Broxton and Zeng, 2014).

2.2.3. Drought index

We used the Standardized Precipitation and Evapotranspiration Index (SPEI), which is a site-specific drought indicator of deviations from the average water balance, to assess the severity and duration of droughts (Vicente-Serrano et al., 2010a, Vicente-Serrano et al., 2010b). SPEI is superior to other drought indices because it is generated by combining temperature and precipitation data (Vicente-Serrano et al., 2010a, Vicente-Serrano et al., 2010b). The monthly SPEI dataset was downloaded from SPEIbase v.2.5 at the Consejo Superior de Investigaciones Científicas (CSIC, <http://digital.csic.es/handle/10261/153475>). The database covers the period between 1901 and 2015 with a spatial resolution of 0.5° and time-scales that range from 1 to 48 months. It is evaluated based on the average water balance (precipitation minus potential evapotranspiration), where potential evapotranspiration is calculated based on the Food and Agriculture Organization – 56 Penman-Monteith method (Vicente-Serrano et al., 2010a, Vicente-Serrano et al., 2010b). For its multi-scalar essence, SPEI can represent the antecedent drought conditions as well as the accumulated values of water deficit. The accumulated water deficit at month j is defined as the sum of the surplus value of SPEI of n months (Peng et al., 2019). For example, the 1-month SPEI and 12-month SPEI provides the drought conditions at the current month and whole year, respectively. In this study, 1-month SPEI to 12-month SPEI datasets from 1982 to 2015 were used to assess cumulative and time-lag effects of drought on the grassland NDVI.

2.3. Methods

2.3.1. Cumulative effects of drought on grassland NDVI

To obtain the explanation of the cumulative effect of drought on the grassland NDVI, Pearson correlation analysis was used to calculate the correlation coefficient determination (r). The results of this analysis can range from -1 to $+1$, which indicate a correlation from negative to positive, respectively. In the current study, Pearson correlation coefficients ($P < 0.05$) of the month NDVI and accumulated SPEI at one-to

twelve-month time scale were considered as candidates for evaluating the cumulative effect of drought on grassland vegetation (Equation (1)). The length of month with maximum coefficients was considered as the cumulative effect of drought on grasslands vegetation (Equation (2)). From this, two important datasets of the cumulative effect of drought on vegetation can be generated: the cumulative time interval (e.g. how many months), and the correlation under the time interval.

$$r_j = \text{corr}(\text{NDVI}, \text{SPEI}_j) \quad 1 \leq j \leq 12 \quad (1)$$

$$r_{\text{max-cml}} = \max_{1 \leq j \leq 12} (r_j) \quad (2)$$

where r_j is the Pearson correlation coefficient between NDVI and SPEI, j is the accumulated time from one-to twelve-month SPEI, and $r_{\text{max-cml}}$ is the maximum value of r_j .

2.3.2. Time-lag effects of drought on grassland vegetation

The time-lag effects of drought on vegetation were also calculated by the Pearson correlation coefficients. Instead of using the cumulative SPEI, only the 1-month SPEI was used to. At each lagged time interval ($0 \leq i \leq 12$), we first calculated the correlation coefficients ($r_0, r_1, r_2, \dots, r_{12}$; Equation (3)) between the NDVI and the 1-month SPEI (SPEI₀, SPEI₁, ..., SPEI₁₂). Following this, the r_i was taken as the optimal correlation when the value reached its maximum correlation coefficient, and the month number (i) is considered as the optimal lagged time (Equation (4)).

$$r_i = \text{corr}(\text{NDVI}, \text{SPEI}_i) \quad 0 \leq i \leq 12 \quad (3)$$

$$r_{\text{max-lag}} = \max(r_i) \quad 0 \leq i \leq 12 \quad (4)$$

where r_i is the Pearson correlation index with a time lag of i months, where i ranges from 0 to 12 (0 indicates no time-lag effects, and 1–12 indicates a 1–12 month time lag, respectively), NDVI represents the month GIMMS3g NDVI time series (1982–2015), SPEI is the month of the time series for 1-month SPEI, and $r_{\text{max-lag}}$ is the maximum value of r_i . The lag month i that has the highest correlation was taken as the best time lag.

Meanwhile, in order to evaluate the response of the cumulative and lagged effects in respect to hydrological conditions, the SPEI-12 in December (annual SPEI) from 1982 to 2015 was selected as a proxy of the annual water balance to evaluate the effect of water balance gradient. The water balance gradient was determined by equal interval values (0.10) based on the mean value of annual SPEI during 1985–2015 (Peng et al., 2019).

3. Results

3.1. Cumulative effect of drought on grassland NDVI

3.1.1. Response of grassland NDVI to drought at different time scales

The correlation between cumulative SPEI and NDVI had large variation across the time scales (Fig. 2). The correlation between NDVI and accumulated SPEI generally increased from one to eight months but then declined, with significant correlation exhibited between 5 and 10 months. The maximum value was found at 8 months ($r = 0.133$, $P < 0.01$), whereas the minimum value was found at 1 month ($r = 0.009$, $P > 0.1$) (Fig. 2). A similar temporal pattern was found for the percent of area which had a positive correlation (Fig. 2). For the correlations between NDVI and the one- and two-month accumulated SPEI, the areas of significantly positive correlations were lower than 6% (Fig. 2), and mostly located in the Inner Mongolia region (Fig. S1a and Fig. S1b). As the accumulated months increased, the area of positive correlation increased as well, peaking at seven (75.64%) and eight months (75.68%), and was found to be distributed in the northwest region of the CLP (Fig. 2, Fig. S1g and Fig. S1h). The positively correlated area dropped to 28.38% with 12 months of accumulated SPEI, and was located across eastern Gansu, northwest Inner Mongolia, and northeast

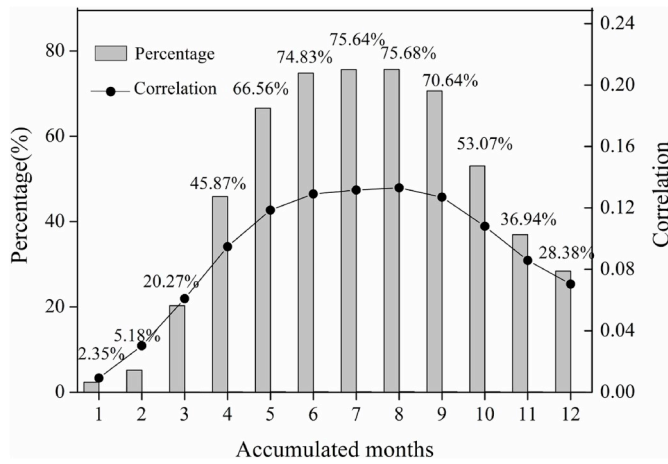


Fig. 2. The spatial distribution of correlation coefficients of grassland NDVI responses to SPEI at different cumulative time scales. The two arrows indicate the ranges that are significant at $P < 0.05$.

Shaanxi (Fig. 2 and Fig. S1). A negative but not significant correlation between NDVI and accumulated SPEI was limited to the area in the Qinghai and Ningxia provinces (Fig. S1a).

3.1.2. Spatial distributions of the cumulative effects of drought on grassland NDVI

Based on the correlation analysis of NDVI with accumulated SPEI, the cumulative effect of SPEI on grassland NDVI was generated using the accumulated month with the largest correlation coefficient ($r_{max-cum}$). Overall, most grassland (80.81%) in the CLP was dominated by significant positive correlations, which were distributed in the eastern Gansu, southern Inner Mongolia, northern Shaanxi, and eastern Qinghai regions (Fig. 3a). The optimum cumulative time interval was primarily from 5 to 9 months, covering 91.95% of grassland in the CLP. In the optimum cumulative time scale, the 6-month and 8-month time scales maintained the largest grassland area (25.50% and 30.01%), including eastern Gansu, the southern Inner Mongolia, northern Shaanxi, and the southern Ningxia (Fig. 3b).

3.1.3. Cumulative effects of drought on grassland NDVI with different water availability conditions

Fig. 4 shows the relationship between the mean $r_{max-cum}$ and corresponding accumulated months, and the mean annual SPEI (water balance gradient). The accumulated months had significantly negative correlation with the mean annual SPEI ($R^2 = 0.90$, $P < 0.001$), indicating that drought had cumulative effects on the grassland NDVI during the longer time scale in the drier regions (Fig. 4a). The mean $r_{max-cum}$ also was significantly negative correlated with the mean annual SPEI ($R^2 =$

0.70, $P < 0.001$), indicating that the grassland NDVI usually experienced stronger cumulative effects of drought in the drier regions (Fig. 4b).

3.2. Time lag effect of drought on grassland vegetation

3.2.1. Response of grasslands NDVI to drought by time lag scales

The time lags of grassland vegetation response to drought in the CLP indicated that most of the grassland responded to drought within 2–3 months. Drought was found to have a variable time-lag effect on vegetation in the CLP. The correlation between NDVI and monthly SPEI increased rapidly from the one-month lag, and reached a peak value at the three-month lag (Fig. 5). It then declined from the lag of three to nine months, but then increased again from nine to 12 months. The NDVI of most of the areas of the CLP was significantly positively correlated with the one-, two-, and three-month SPEI, indicating a quick response of vegetation to the drought. In detail, 10.37% of the area had a significant positive correlation between NDVI and the one-month SPEI. This then increased to 35.94% with the two-month SPEI, and reached 46.12% with the three-month SPEI. These areas are mainly located in the western Inner Mongolia, eastern Ningxia, and northeast Shaanxi regions (Figs. S2a–c).

3.2.2. Spatial distributions of the time-lag effects of drought on grassland NDVI

With the maximum correlation ($r_{max-lag}$) between NDVI and one-month SPEI (a time-lag range from 0 to 12 months), the time-lag effect was summarized for each pixel. Overall, 54.52% of the grassland presented a time-lag response to drought, distributed in the western Inner Mongolia, eastern Ningxia, and northeast Shaanxi regions (Fig. 6a). Among the area with a time-lag response, about 96.85% maintained a one-to three-month time-lag response of vegetation to drought, and the $r_{max-lag}$ at time-lags of two-to three-months occupied most of the area (33.35% and 56.17%; Fig. 6b).

3.2.3. Time lag effects of drought on grassland NDVI with the different water availability conditions

The patterns of the lagged effect of drought on grassland NDVI with different water availability conditions were calculated (Fig. 7). Negative correlations between the mean annual SPEI (water balance gradient), mean $r_{max-lag}$, and corresponding mean lagged months were found. In the view of lagged time scale, the mean lagged months indicated a decreased trend (from 3.14 to 2.86) with an increasing mean annual SPEI, indicating that drought had increased lagged effects on the grassland NDVI within longer time scales in the drier areas when compared to wetter areas ($R^2 = 0.547$, $P < 0.05$). From the perspective of $r_{max-lag}$, the mean $r_{max-lag}$ showed a significant negative correlation to the mean annual SPEI ($R^2 = 0.785$, $P < 0.01$), indicating that the grassland NDVI in arid areas usually experienced a stronger time-lag effect (Fig. 7b).

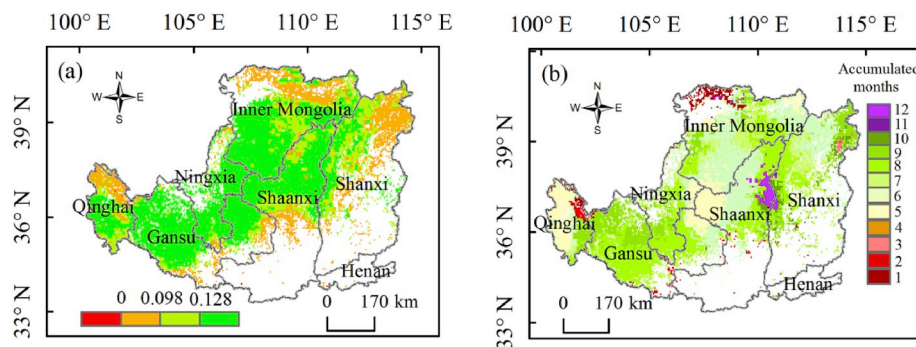


Fig. 3. (a) Spatial pattern of the $r_{max-cum}$ between SPEI and NDVI during the period of 1982–2015. (b) Spatial distribution of the corresponding accumulated months when the $r_{max-cum}$ occurred.

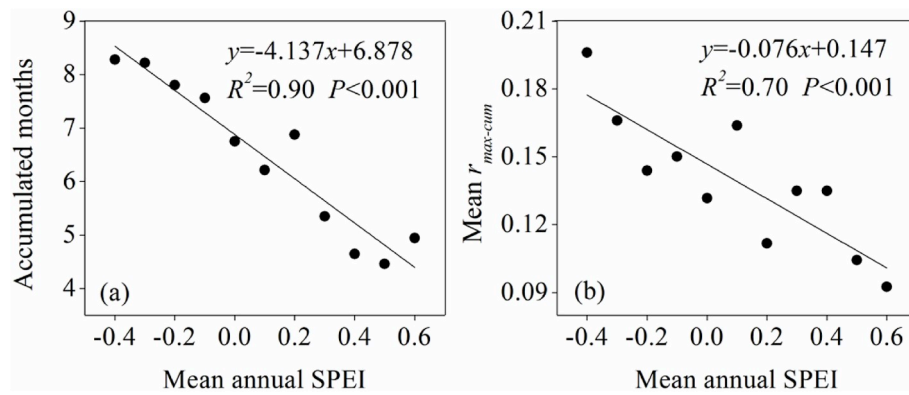


Fig. 4. (a) Relationships between accumulated months and mean annual SPEI (b) Relationships between the mean.

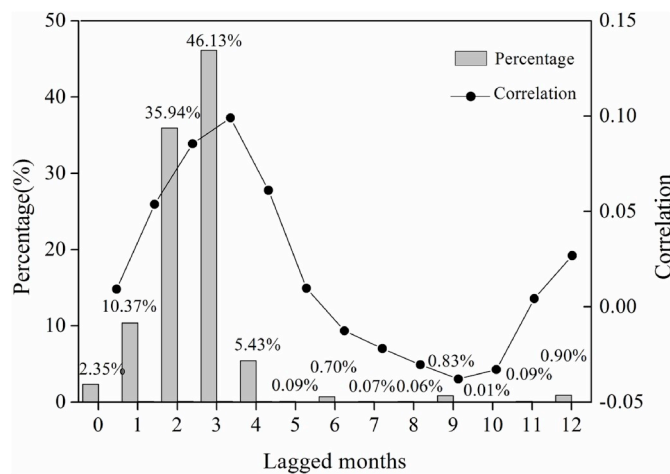


Fig. 5. Correlations and area percentage of significant correlations ($P < 0.05$) between 1-month SPEI and grassland vegetation with a time lag of 0–12 months.

4. Discussion

4.1. The cumulative effect of drought on grassland vegetation

Vegetation growth is greatly impacted by water deficits. The CLP is a critical ecological constructive area where grassland is one of the most important and dominant vegetation types (Gang et al., 2018; Zheng et al., 2019). However, the CLP is suffering from severe droughts which can have significant impacts on the ecosystem within the region. Zhao

et al. (2019) reported that 21.54% of the CLP was affected by drought. Hua et al. (2017) confirmed that vegetation activity during the growing season was highly correlated with drought occurrences in the semi-arid regions of China. The results of this study indicate that cumulative water deficits affect vegetation growth in most of the grassland (80.84%) in the CLP, over a quite long cumulative time scale of five-to nine-months (Fig. 2). In the arid and semi-arid areas, the grassland has widely developed with well-developed root systems, and as soil moisture is the primary source of water for vegetation growth (Guo and Shao, 2002), water availability at longer timescales may have a significant influence on vegetation activity (Fig. 4), which is a result supported by several previous studies (Ji and Peters, 2003; Hua et al., 2017).

4.2. The time lag effect of drought on grassland vegetation

Similar to the cumulative effect, vegetation in a large proportion of the grassland in the CLP has a time-lag response to droughts (Fig. 5). Most of the grassland vegetation was found to have a one-to three-month time-lag to droughts. Similar to the time-lag effect of precipitation at the end of growing season (Cong et al., 2017; Peng et al., 2019), our study results indicated that the three-month drought lag had the strongest impact on grassland NDVI. In the arid, semiarid, and semi humid areas, the major limiting factor for vegetation growth was soil moisture, and a two-to three-month lag was found in the responses of grassland NDVI in these areas to drought. This may be due to the intrinsic physiological mechanisms of vegetation that allow the species in these areas to tolerate water deficits for longer periods of time, and similar results were also found in previous studies (Anderegg et al., 2015; Vicente-Serrano et al., 2013; Wen et al., 2019). The water balance gradient was significantly correlated with the mean $r_{max-lag}$ and lagged months. The value of the mean $r_{max-lag}$ was significantly negatively correlated with the water

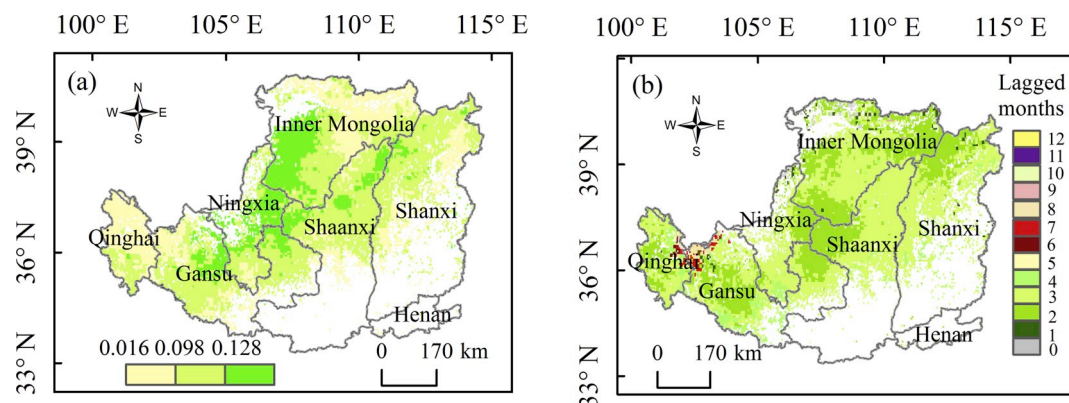


Fig. 6. (a) Spatial distributions of the $r_{max-lag}$ between 1-month SPEI and NDVI during the period of 1982–2015. (b) Spatial distribution of the corresponding lagged months where the $r_{max-lag}$ occurred.

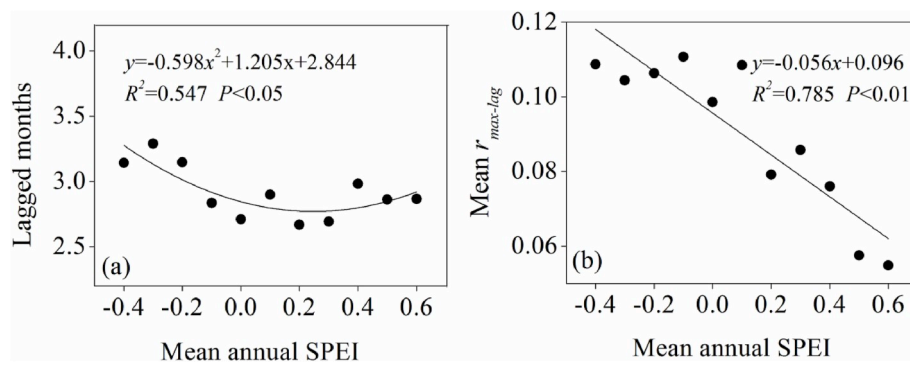


Fig. 7. (a) Relationships of between lagged months and mean annual SPEI (b) Relationships between the mean.

balance gradient, indicating that drought had a stronger impact on grassland NDVI in drier areas than in wetter areas (Fig. 7b). The grassland NDVI usually responds to drought at longer lagged months in drier regions (lower mean annual SPEI) (Fig. 7a). The grasslands of CLP are located in the arid and semiarid region, and the moisture is the principal limiting factor of vegetation growth (Wen et al., 2019). Due to the physiological mechanisms of vegetation, the vegetation in the arid have stronger adaptability to moisture shortage than vegetation in the semi-arid have abilities. Similar results were also reported in previous studies.

4.3. Limitations and implications for future work

Some limitations and uncertainties still exist in this study. First, NDVI and SPEI were used to analyze the responses of grasslands growth to drought, and it should be noted that the SPEI values were calculated by using the monthly precipitation and potential evapotranspiration. While the dynamics of precipitation and potential evapotranspiration are also significant characteristics of climatic condition, the impacts of their dynamics on vegetation growth can provide important insights into the mechanism of vegetation response to climate. Numerous studies have reported that increasing potential evapotranspiration seems to intensify the aridity of an environment (Cook et al., 2014). Although the precipitation and potential evapotranspiration were considered to calculate SPEI in this study, how the complex interactions between different environmental factors affect vegetation growth requires more studies. Furthermore, the correlation analysis was adopted to calculate the relationships between grassland vegetation and drought. However, the relationships between vegetation growth and drought can be nonlinear due to their complicated interactions involved. Therefore, using simple correlation analysis may generate oversimplified results about the cumulative and time-lag effects of vegetation's response to drought (Liu et al., 2015; Wen et al., 2019). Lastly, this study focused only the responses of vegetation to drought, while the impacts from human activities (such as ecological recovery programs, urban sprawl and agricultural irrigation), other no-climatic factors (such as CO₂ fertilization, nitrogen and phosphorus deposition), and natural disturbances (such as wildfires, plant diseases and insect pests) can also have unneglectable impacts on vegetation growth (Flower and Gonzalez-Meler, 2015; Huang et al., 2018; Liu et al., 2017). Efficient management of vegetation would be facilitated by comprehensive understanding of how vegetation responds to different types of disturbance.

5. Conclusions

In the present study, using the vegetation index (NDVI) and monthly SPEI, we assessed the cumulative and time-lag effects of drought on grassland growth over in the period of 1982–2015. The key results show that: antecedent water deficits cumulatively impact vegetation

variations; drought had a quite long cumulative effect (5- to 10- month) on vegetation growth in most of the grassland of the CLP. The cumulative effects of drought on grassland NDVI were presented in 80.81% of the grassland in the CLP, mostly located in the eastern Gansu, southern Inner Mongolia, northern Shaanxi, and southern Ningxia regions. Vegetation in arid areas was found to experience stronger cumulative effects of drought which were characterized by a longer cumulative time scale, and the time-lag effect of drought on vegetation growth in the study area mainly happens in one to three months, with the three-month lag having the strongest effect on grassland vegetation. Vegetation in about half of the grassland in the CLP was found to have experienced a time-lag effect of drought, primarily in areas located in the western Inner Mongolia, eastern Ningxia, and northeast Shaanxi regions. The lagged months and mean r_{max_cum} had a negative correlation with the mean annual SPEI, respectively ($R^2 = 0.547$, $P < 0.05$; $R^2 = 0.785$, $P < 0.01$). Overall, the drought events had significant cumulative and time-lag effect on grassland growth, this study provides a comprehensive understanding of the cumulative and time-lag effects of drought on vegetation growth in the grassland of the CLP, and the results hold significant value for vegetation conservation strategies, response of vegetation to climate change, and water and soil restoration in the CLP. Our study highlights the importance of both cumulative and lag effect of drought on vegetation. Due to vegetation's different lag and cumulative response to drought, planners should consider what type of vegetation to plant and how to conserve existing vegetation.

CRedit authorship contribution statement

Anzhou Zhao: Conceptualization, Methodology, Writing - original draft, Visualization. **Qiuyan Yu:** Methodology, Writing - review & editing. **Lili Feng:** Visualization, Investigation. **Anbing Zhang:** Data curation, Validation. **Tao Pei:** Writing - review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2020.110214>.

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